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Addressing semantic heterogeneity through multiple knowledge base assisted merging of domain-specific ontologies

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ABSTRACT

With the development of the Semantic Web (SW), the creation of ontologies to formally conceptualize our understanding of various domains has widely increased in number. However, the conceptual and terminological differences (a.k.a *semantic heterogeneity problem*) between ontologies form a major limiting factor towards their use/reuse and full adoption in practical settings. A key solution to addressing this problem can be through identifying semantic correspondences between the entities (including concepts, relations, and instances) of heterogeneous ontologies, and consequently achieving interoperability between them. This process is also known as *ontology alignment*. The output of this process can be further exploited to merge ontologies into a single coherent ontology. Indeed, this is widely regarded as a crucial, yet difficult task, specifically when dealing with heavyweight ontologies that consist of hundreds of thousands of concepts. To address this issue, various ontology merging approaches have been proposed. These approaches can be classified into three categories: single-strategy-based approaches, multiple-strategy-based approaches, and approaches based on exploiting external semantic resources. In this paper, we first discuss the strengths and limitations of each of these approaches, and then present our framework for addressing the semantic heterogeneity problem through merging domain-specific ontologies based on multiple external semantic resources. The novelty of the proposed approach is mainly based on employing knowledge represented by multiple external resources (knowledge bases in our work) to make aggregated decisions on the semantic correspondences between the entities of heterogeneous ontologies. Other important issues that we attempt to tackle in the proposed framework are: (i) Identifying and handling inconsistency of semantic relations between the ontology concepts and, (ii) Handling the issue of missing background knowledge (such as concepts and instances) in the exploited knowledge bases by utilizing an integrated statistical and semantic technique. Additionally, the proposed solution soundly enriches the knowledge bases with missing background knowledge, and thus enables the reuse of the newly obtained knowledge in future ontology merging tasks. To validate our proposal, we tested the framework using the OAEI 2009 benchmark and compared the produced results with state-of-the-art syntactic and semantic based systems. In addition, we utilized the proposed techniques to merge three heavyweight ontologies from the environmental domain.

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1. Introduction

The incorporation of semantic technology in information systems is regarded as an important issue, particularly with the development of Web 3.0. The semantics are captured in domain-specific ontologies, which aim at providing a formal, explicit and shared conceptualization and understanding of common domains between different communities [19,38]. With the advent of the Internet, which has enabled the development of an ever-increasing number of ontologies with different terminologies, it has become difficult to make use of this vast and heterogeneous source of knowledge. The difficulty of this task is due to the decentralized nature of ontology development and the differences between the viewpoints of ontology engineers. This has resulted in the so called "semantic heterogeneity" problem, which constitutes the major obstacle against achieving interoperability between ontologies.

Solving the semantic heterogeneity problem can be achieved through merging two or more ontologies from the same domain

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into a single coherent ontology [34]. Several automatic and semi-automatic ontology merging approaches have been proposed. Details on the types of strategies that are used by state-of-the-art ontology merging systems are listed below:

- Strategies based on linguistic matching a.k.a. Name-based strategies: these approaches compute distances between the strings of the concepts (e.g. Jaro-Winkler distance function [43]) from the source ontologies to obtain correspondences between them [5,8]. However, they do not take the semantic aspects of the compared strings into account and therefore, errors like considering the concepts "car" and "care" as "equivalent" concepts, or considering the concepts "car" and "automobile" as "not equivalent" concepts would be produced.
- Structure-based similarity strategies: these approaches rely on the structure of the source ontologies to merge them [1,10,12,33]. Typically, the graph structure of both ontologies is provided (through is-a or other relations) and the similarity is computed for each concept based on its neighboring concepts. In this context, the neighbors of each concept are its parents (ancestors) and/or its children (descendants). However, when the number of concepts is huge such as in heavyweight ontologies, these approaches suffer from resource consumption problems as they utilize in-memory structures to merge both ontologies [45].
- Combination-based strategies: other approaches rely on the combination of name-based and structure-based similarity measures such as in the systems proposed by the authors of [10,29,30,42]. Although these techniques may produce good results, there is a considerable number of cases where they fail to discover semantic correspondences between the source ontologies due to the drawbacks of each individual strategy discussed above.
- Strategies based on external auxiliary resources: these approaches propose to integrate an external resource or knowledge base to support the task of finding semantic correspondences between heterogeneous ontologies [7,17,27,41]. However, these approaches are subjected to the limitations of the exploited knowledge base. For example, Aleksovski et al. use a background knowledge based paradigm in the medical domain where the DICE ontology acts as a semantic bridge between the matched ontologies [2]. Also, Sabou et.al. provide an approach to ontology matching which can exploit multiple heterogeneous ontologies obtained from the Web [37]. Another example is the system proposed by the authors of [46]. This system employs knowledge represented by an external resource (WordNet [32] and Web pattern-based queries to derive the semantic aspects of the entities of the source ontologies. However, the most successful and widely employed knowledge bases (e.g. WordNet, Cyc, OpenCyc, SUMO, etc.) are man-made; they suffer from low coverage, high assembly cost and fast aging whereby they do not know the latest Windows version or latest soccer stars [40]. Typically, background information such as concepts and instances are missing. For example, we find that concepts such as "Corporate Body" or instances such as "Monash University" are missing when using WordNet as an external knowledge base.

In this paper, we propose an ontology merging framework that takes two domain-specific ontologies as input, finds semantic correspondences (alignments) between both ontologies and produces a single merged ontology as output. In our approach, decisions on the semantic correspondences between the entities of semantically heterogeneous ontologies are made by considering multiple knowledge bases. When the exploited knowledge bases have missing background knowledge such as concepts or instances, we utilize other techniques to capture their semantics. To do this, we employ a process integrating name-based and coupled statistical and semantic based techniques respectively based on the Jaro-Winkler distance and the Normalized Retrieval Distance (NRD) functions. It is important to mention that other string edit distance techniques can be used; however, we found the Jaro-Winkler function among the best techniques to compute the string distances between the labels of the ontologies' entities. Moreover, the implementation of this function is publically available and can be easily integrated to any framework that deals with string processing. This step has three major benefits. First, it tackles the issue of the single use of string distance to obtain correspondences between ontology concepts, such as in name-based approaches, by considering an additional coupled statistical and semantic technique. Then, it enables the reuse of newly obtained knowledge in subsequent ontology merging. Finally, it eliminates the concern of manually defining the relations between the missing concepts or instances and other concepts of the knowledge bases. Accordingly, we summarize the contributions of the paper as follows:

- Prioritizing the ontology merging techniques into semantic-based, name-based, and statistical-based techniques respectively.
- Exploiting multiple knowledge bases to make aggregated mapping decisions for merging heterogeneous ontologies.
- Addressing the issue of missing background knowledge in the exploited knowledge bases by utilizing a process integrating name-based and coupled statistical and semantic-based techniques.
- Enriching knowledge bases based on the obtained information. It is important to mention that although some of the approaches use external resources such as knowledge bases to support the merging task, they use only one knowledge base and do not attempt to enrich the knowledge base with missing background knowledge as we do in our framework.

The rest of this paper is organized as follows. Section 2 gives a general overview of the proposed framework. Background information related to the proposed framework is presented in Section 3. We detail the steps of inconsistency checking and resolution between heterogeneous domain specific ontologies and knowledge bases in Section 4.1. The multiple knowledge base assisted merging process is developed in Section 4.2 while dealing with missing background knowledge is discussed in Section 4.3. We develop knowledge base enrichment in Section 4.4. Section 5 presents the experimental results carried out to evaluate the effectiveness of the employed methods and techniques in the proposed framework. The final section presents the conclusions and outlines the future work.

2. General overview

The first technique that we utilize to find semantic correspondences between the entities of different ontologies is based on employing multiple knowledge bases. Each knowledge base represents a repository of facts about entities and their relationships that exist in different domains. In this context, a fact is a triple consisting of an *entity-relation-entity* structure, where entities are related through different types of semantic and taxonomic relations. These relations are automatically extracted from multiple heterogeneous data sources such as plain texts, image and video captions and online ontologies. It is important to mention that

knowledge base construction and maintenance has always been at the heart of the Semantic Web technology. Apparently, with the continuous expansion of the Web, this task has become increasingly difficult [18]. To address this issue, several knowledge base construction systems have been proposed [3,14,21,39]. Some of these systems rely on human input to enrich the knowledge base and keep it up-to-date. Examples of produced knowledge bases by such systems are Freebase[1] and True Knowledge.[2] On the other hand, other systems rely on a single data source to create a knowledge base or multiple knowledge bases that are related to a particular domain. For instance, Geifman and Rubin, propose to model and store knowledge about age-related phenotypic patterns and events in an Age-Phenome Knowledge Base (APK) [16]. Another example is the terrorism knowledge base, which contains all relevant knowledge about terrorist groups, their members, leaders, affiliations, and full descriptions of specific terrorist events [11]. This knowledge base was integrated into Cyc [28], which is a general-purpose ontology that captures knowledge from multiple domains. In our approach, we not only aim at avoiding the effort required by users to manually maintain and update the knowledge base, but also at populating and extending knowledge bases through a coupled statistical and semantic technique.

In order to illustrate the proposed framework, we take an example of two domain-specific ontologies, Biblio.owl and BibTex.owl that are used to describe the Bibliography domain. Fig. 1 shows the class (a.k.a. concept) hierarchy of both ontologies. Classes in each ontology are connected through the "is-a" partial order. An initial check on each of the input ontologies is performed to validate whether the semantic relations between the concepts of the input ontologies are contradicting the semantic relations that are used to connect the same concepts in the knowledge bases. In any case of contradiction, we resolve it through reconstructing the hierarchy of the input ontology based on referring to the knowledge bases.

For example, we find that the concept "Agent" is a hypernym of the concept "Person" in the Biblio ontology, while in the knowledge bases KB_1 and KB_2, "Agent" is a hyponym of "Person". This conflict is resolved based on a voting decision made by considering knowledge bases KB_1, KB_2, . . . , KB_*n*. Next, we reconstruct the hierarchy of the Biblio ontology by taking into account the semantic information obtained from the knowledge bases. The same process applies to the BibTex ontology. After resolving the semantic conflicts, different heuristics are applied to merge both ontologies through finding semantic relations between their entities. In our approach, the semantics-based ontology merging strategy is given priority on top of other techniques.

However, a problem may arise when trying to find the semantic relation between two concepts or instances while one or both of them are not defined in the knowledge bases (i.e. missing knowledge). In such case, we utilize a name-based technique to measure the distance between the strings of the concepts or instances in the ontologies. This technique is only employed to find equivalent concepts or instances. For other types of mapping relations, we exploit another external resource, the World Wide Web (WWW), in order to measure the semantic relatedness between the missing concept or instance and other concepts and instances in the merged ontology. Then, we rely on the obtained information about the missing concepts or instances to derive their semantics. Therefore, we first start by employing the name-based technique to obtain, if any, a set of equivalent concepts to this concept. Then, we utilize the statistical technique to find other types of relations that may exist between this concept and the other concepts. Based on the obtained statistical results, we re-locate the missing concept in the hierarchy of the merged ontology.

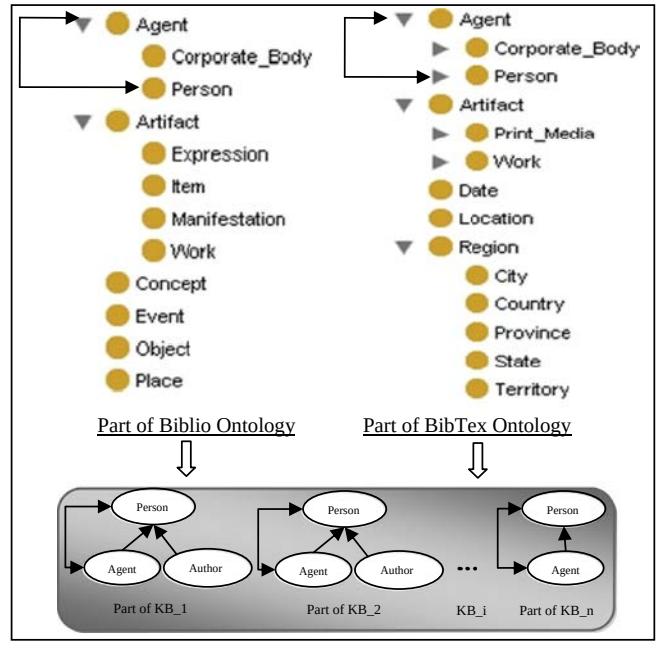

Fig. 1. Contradiction between the knowledge bases and the input ontologies.

The final step consists in enriching the knowledge bases with the missing background knowledge. Enriching a knowledge base with a new concept or instance requires finding the appropriate path(s) for locating it. To do this, we compute the path similarity between the merged ontology and the path(s) in the knowledge bases. We do a semantic-based comparison between the path(s) that originate(s) from the missing concept in the merged ontology and map it/them to the right path(s) in the knowledge bases.

3. Background

Before we proceed to presenting the details of the used methods and techniques, we introduce in the context of our framework the terms "Domain-specific Ontology", "Knowledge Base", "Alignment", "Merging", "Enrichment", and "Normalized Retrieval Distance".

Definition 1. A **domain-specific ontology** Ω is a 4-tuple $\langle C, R, I, A\rangle$ where:

- $C = \{(c_i), i \in [1, |C|]\}$ represents the set of domain concepts of the ontology. The concept hierarchy of Ω is a pair $(C, \leqslant)$, where $\leqslant$ is an order relation on $C \times C$.
- $R = \{(r_i), i \in [1, |R|]\}$ represents the set of semantic relations holding between the ontology concepts (example relations are presented in Section 4).
- I is the set of instances or individuals.
- A is the set of axioms verifying: $A = \{(r_i, c_j, c_k)\}$ s.t. $i \in [1, |R|]$, $j, k \in [1, |C|]$, $c_j, c_k \in C$ and $r_i \in R$.

Definition 2. A **knowledge base** KB is an ontology covering several heterogeneous semantic domains and is defined as a 4-tuple $(C_{KB}, R_{KB}, I_{KB}, A_{KB})$ where:

[1] http://www.freebase.com/.
[2] http://www.trueknowledge.com/.

- C_{KB} is the set of concepts that are defined in KB. Since KB covers information from multiple domains and is not limited to a particular domain, we have $|C_{KB}| \gg |C|$. However, some concepts of domain-specific ontologies may be missing in KB. This illustrates the missing background knowledge issue.
- R_{KB} is the set of semantic and taxonomic relations that are defined to relate the set of concepts C_{KB}.
- I_{KB} is the set of instances of the concepts that are defined in KB.
- A_{KB} is the set of axioms verifying: $A_{KB} = \{(r_i, c_j, c_k)\}$ s.t. $i \in [1, |R_{KB}|]$, $j, k \in [1, |C_{KB}|]$, $c_j, c_k \in C_{KB}$ and $r_i \in R_{KB}$.

Definition 3 (*Alignment*). Is the set of corresponding entities between two or more ontologies. The alignment is the output of the matching process [15].

Definition 4. Given two domain-specific ontologies $\Omega 1$ and $\Omega 2$, the **merging** operation finds semantic correspondences between their concepts and produces a single merged ontology Ω_{merged} as output. Semantic correspondences between both ontologies are 4-tuples $\langle c_{id}, c_i, c_j, r\rangle$ such that:

- c_{id} is a unique identifier of the identified correspondence.
- $c_i \in \Omega 1$, $c_j \in \Omega 2$ are corresponding concepts of the input ontologies.
- $r \in R$ is a semantic relation holding between both entities c_i and c_j.

Definition 5 (*Enrichment*). An ontology **enrichment** process takes a text corpus ζ and an ontology Ω as input and produces, for each entity $c \in C$, a set $S(c) \subseteq C(\zeta)$ where:

- $S(c)$ is the set of suggested enrichment candidates for c. A suggested candidate $c_\zeta \in C(\zeta)$ is a concept extracted from ζ
- $C(\zeta)$ is the set of concepts from ζ

The set $S(c)$ is obtained by using the NRD function defined below and depends on a threshold value v according to Eq. (1).

$$S(c, v) = \{c_\zeta \in C(\zeta) | \mathrm{NRD}(c_\zeta, c) \geqslant v\} \tag{1}$$

Definition 6 (*Normalized Retrieval Distance (NRD)*). Is an adapted form of the Normalized Google Distance (NGD) [6] function that measure the semantic relatedness between pairs of entities (such as concepts or instances): Given two entities *e_mis* and *e_in*, the Normalized Retrieval Distance between *e_mis* and *e_in* can be obtained as follows:

$$\mathrm{NRD}(e_mis, e_in) = \frac{\max\{\log f(e_mis), \log f(e_in)\} \log f(e_mis, \mathrm{e}_in)}{\log\ M - \min\{\log f(e_mis), \log f(e_in)\}} \tag{2}$$

where

- *e_mis* is an entity that is not defined in the ontology,
- *e_in* is an entity that exists in the ontology,
- *f(e_mis)* is the number of hits for the search entity *e_mis*,
- *f(e_in)* is the number of hits for the search entity *e_in*,
- *f(e_mis, e_in)* is the number of hits for the search entities *e_mis* and *e_in*,
- *M* is the number of indexed Web pages considered.

This computation is necessary to examine how closely related two entities are by analyzing pairwise co-occurrence frequencies. A distance of zero indicates that they always appear together. Formally, this is a measure for the symmetric conditional probability of co-occurrence of *e_mis* and *e_in*. Given a document containing one of *e_mis* or *e_in*, NRD(*e_mis*, *e_in*) measures the probability of that document also containing the other concept or instance.

4. Detailed steps of the proposed framework

4.1. Inconsistency checking and resolution

At this step, for each of the input ontologies, we attempt to find existing inconsistencies in the semantic relations between its concepts and those in the knowledge bases. To do this, we use the latter for computing the semantic relations between the input ontology concepts. These semantic relations are:

- **Equivalence** ($\equiv$): Either concepts are equal or one of them is a synonym of the other (e.g. Student is equivalent to Pupil).
- **Specialization** ($\subset$): If a concept or one of its synonyms is a hyponym or meronym of another concept or its synonyms (e.g. Student is less general than Person).
- **Generalization** ($\supset$): If a concept or one of its synonyms is a hypernym or holonym of another concept or its synonyms (e.g. Transport is more general than Car).
- **Disjointness** ($\perp$): If either concepts or their synonyms are antonyms or different hyponyms of the same synset (e.g. Book is disjoint from Phone).
- **Unknown** relation (??): If one or both of the compared concepts or instances is/are missing in the knowledge bases. This relation tells us that the exploited knowledge bases have missing background knowledge and need enrichment.

For example, Table 1 shows some of the identified relations between the concepts of both the Biblio and BibTex ontologies.

To resolve the semantic inconsistencies that exist between the relations that hold between the concepts of the input domain-specific ontologies and those that hold between the same concepts which are defined in the used knowledge bases, we consider the following cases:

4.1.1. One-to-one

This case represents having a semantic relation between two concepts in the domain-specific ontology that is inconsistent with the semantic relation between the same concepts in only one of the knowledge bases. In this case, we believe that the semantic relation in the domain-specific ontology should be given more weight than the one defined in the knowledge base. The reason behind our choice lies in the fact that domain-specific ontologies are usually created by communities to deeply capture their knowledge in a reusable form. We therefore suggest that they are in this case more accurate than the knowledge base, which usually provides a broad definition of the concepts that are used to describe multiple domains of interest. Therefore, we retain the semantic relation in the domain-specific ontology.

4.1.2. One-to-many

This scenario represents having a semantic relation between two concepts in the domain-specific ontology that is contradicting the semantic relation between the same concepts in more than one knowledge base. In this case, we believe that the semantic relation in the domain-specific ontology should be changed according to the one defined in the knowledge bases. This decision is justified by the agreement found in this case between different domain experts who built the knowledge bases.

In order to illustrate this case, we consider the following example. In Fig. 1, the concept "Agent" is considered as a super-concept of "Person" in the Biblio ontology. Below is the OWL syntax:

```
<?xml version=''1.0''?>
<rdf:RDF
    xmlns:xsp=''http://www.owl-ontologies.com/
  2005/08/07/xsp.owl#''
    xmlns:swrlb=''http://www.w3.org/2003/11/
  swrlb#''
    xmlns:swrl=''http://www.w3.org/2003/11/
  swrl#''
    xmlns:protege=''http://protege.stanford.edu/
  plugins/owl/protege#''
    xmlns:rdf=''http://www.w3.org/1999/02/22-rdf-
  syntax-ns#''
    xmlns:xsd=''http://www.w3.org/2001/
  XMLSchema#''
    xmlns:rdfs=''http://www.w3.org/2000/01/rdf-
  schema#''
xmlns:owl=''http://www.w3.org/2002/07/owl#''
    xmlns=''http://resc.ai.toronto.edu:8080/
  maponto/Biblio#''
  xml:base=''http://resc.ai.toronto.edu:8080/
  maponto/Biblio''>
<owl:Ontology rdf:about='''' >
<rdfs:comment rdf:datatype=''http://www.w3.org/
  2001/XMLSchema#string''>A bibliographic
  ontology for the form and content of
  bibliographic descriptions from the viewpoint of
  libraries. See the publication of a set of
  cataloguing principles by the International
  Federation of Library Association and
  Institutions (IFLA).</rdfs:comment>
</owl:Ontology>
<owl:Class rdf:ID=''Object''/>
<owl:Class rdf:ID=''Concept''/>
<owl:Class rdf:ID=''Person''>
<rdfs:subClassOf>
    <owl:Class rdf:ID=''Agent''/>
</rdfs:subClassOf>
```

However; "Agent" is considered as a sub-concept of "Person" according to the knowledge bases KB_1 and KB_2 (e.g. WordNet and Yago). Therefore, we consider the semantic relation defined in the knowledge bases KB_1 and KB_2 to supersede the one defined in the Biblio ontology. Therefore, in the latter, we substitute the relation between "Agent" and "Person" with the one given by KB_1 and KB_2. Accordingly, we consider "Agent" as a sub-concept of "Person" and modify the OWL syntax of the Biblio ontology as follows:

```
<?xml version=''1.0''?>
<rdf:RDF
    xmlns:xsp=''http://www.owl-ontologies.com/
  2005/08/07/xsp.owl#''
    xmlns:swrlb=''http://www.w3.org/2003/11/
  swrlb#''
    xmlns:swrl=''http://www.w3.org/2003/11/
  swrl#''
    xmlns:protege=''http://protege.stanford.edu/
  plugins/owl/protege#''
    xmlns:rdf=''http://www.w3.org/1999/02/22-rdf-
  syntax-ns#''
    xmlns:xsd=''http://www.w3.org/2001/
  XMLSchema#''
xmlns:rdfs=''http://www.w3.org/2000/01/rdf-
  schema#''
    xmlns:owl=''http://www.w3.org/2002/07/owl#''
    xmlns=''http://resc.ai.toronto.edu:8080/
  maponto/Biblio#''
xml:base=''http://resc.ai.toronto.edu:8080/
  maponto/Biblio''>
<owl:Ontology rdf:about='''' >
<rdfs:comment rdf:datatype=''http://www.w3.org/
  2001/XMLSchema#string''>A bibliographic
  ontology for the form and content of
  bibliographic descriptions from the viewpoint of
  libraries. See the publication of a set of
  cataloguing principles by the International
  Federation of Library Association and
  Institutions (IFLA).</rdfs:comment>
</owl:Ontology>
<owl:Class rdf:ID=''Object''/>
<owl:Class rdf:ID=''Concept''/>
<owl:Class rdf:ID=''Agent''>
<rdfs:subClassOf>
    <owl:Class rdf:ID=''Person''/>
</rdfs:subClassOf>
```

4.1.3. Majority voting

A conflict may occur between more than two knowledge bases when attempting to decide on the semantic relation between two concepts in the domain-specific ontology. In this case, a voting function is employed to reach a consensual decision and the majority relation is chosen to relate both concepts. In the case of a tie where several knowledge bases each suggest a different solution (e.g. one knowledge base proposes that two concepts are related by the "is-a" relation while another one proposes that they are related by the "part-of" relation), we retain the original relation of the domain-specific ontology. In the case of ties where two or more parties consisting of several knowledge bases suggest different solutions (e.g. three knowledge bases propose that two concepts are related by the "is-a" relation while three other propose that they are related by the "part-of" relation), we retain one solution arbitrarily. It may be argued that it is a difficult task to find consensus among all knowledge bases on the types of semantic relations that might relate two concepts. It is indeed a major limiting factor highlighting the semantic heterogeneity issue. However, in the effort to construct coherent domain-specific ontologies, we believe that a large number of external semantic resources shall

be used to support and enrich the process unlike the conventional single knowledge-based ontology merging approaches.

Algorithm 1 below shows the steps of resolving the contradictions between the domain-specific ontologies and the knowledge bases.

Algorithm 1. Contradiction resolution

Input: domain specific ontology Ω
Output: updated domain specific ontology Ω
1: String relΩ
2: String suggestedRel
3: String[] relKB
4: for each $c_i \in \Omega$
5: for each $c_j \in \Omega$
6: if(isRelated(c_i, c_j, Ω)
7: relΩ = getRelation(c_i, c_j, Ω)
8: for each KB_i
9: relKB[i] = getRelation(c_i, c_j, KB_i)
10: end for
11: if(!isEmpty(relKB))
12: suggestedRel = voteForRelation(relΩ, relKB)
13: replace (relΩ, c_i, c_j) with (suggestedRel, c_i, c_j)
14: end if
15: end if
16: end for
17: end for

For each related pair of nodes in the input ontology, we find the relation that holds between them. The **voteForRelation** function (line 12) takes the relation defined between the node pairs from the input ontology and the relation(s) between the same node pairs from the knowledge bases. Then, the function resolves the conflict, if any, between these relations by considering the cases discussed in sections 4.1.1, 4.1.2 and 4.1.3. Finally, the hierarchy of the input ontology is updated.

4.2. Multiple knowledge base assisted ontology merging

Constructing the hierarchy of the merged ontology Ω_{merged} is based on the identified semantic relations between the concepts of the input domain-specific ontologies $\Omega 1 = \langle C_1, R_1, I_1, A_1 \rangle$ and $\Omega 2 = \langle C_2, R_2, I_2, A_2 \rangle$.

At this step, the concepts (and corresponding instances) that do not exist in the knowledge bases are located in the merged ontology according to their locations in the input ontologies. For the others, decisions for identifying the semantic relations that may hold between them are made by considering the knowledge bases. Since different knowledge bases may identify different or contradictory relations, a voting process inspired from the one presented in section 4.1.3 is considered to handle the inconsistencies that may arise between them Let R_K be the set of relations from the knowledge bases used to relate the concepts of the merged ontology. R is a subset of $\cup R_{KB_i}$ (i.e. the union of the sets of relations from each knowledge base KB_i). Let A_K be the set of axioms from the knowledge bases used to construct the merged ontology. A is a subset of $\cup A_{KB_i}$ (i.e. the union of the sets of axioms pertaining to each knowledge base KB_i). The output merged ontology Ω_{merged} is defined as follows:

$\Omega_{merged} = \langle C_{merged}, R_{merged}, I_{merged}, A_{merged} \rangle$ where:

- $C_{merged} = \{C_1 \cup C_2\}$ represents the set of concepts of the merged ontology including those which are missing in the knowledge bases.
- $R_{merged} = \{R_1 \cup R_2 \cup R_K\}$ represents the set of semantic relations holding between the concepts of the merged ontology.
- $I_{merged} = \{I_1 \cup I_2\}$ is the set of instances or individuals defined in Ω_{merged}.
- A_{merged} is the set of axioms defined in Ω_{merged}. $A_{merged} = \{(r_i, c_j, c_k)\}$ s.t. $i \in [1, \text{Card}(R_{merged})]$, $j, k \in [1, \text{Card}(C_{merged})]$, $c_j, c_k \in C_{merged}$, $r_i \in R_{merged}$ and $(r_i, c_j, c_k) \in A_K \cup A_1 \cup A_2$.

Finally, the hierarchy of the merged ontology is constructed based on the computed semantic relations between the concepts of the input ontologies. Algorithm 2 demonstrates the process of merging the input domain-specific ontologies based on the exploited knowledge bases.

Algorithm 2. Merging ontologies

Input: Two domain-specific ontologies $\Omega 1$ and $\Omega 2$
Output: Merged ontology Ω_{merged}
1: String[][] suggestedRel
2: String[] relKB
3: for each concept $c_i \in \Omega 1$
4: for each concept $c_j \in \Omega 2$
5: for each KB_i
6: if(isRelated(c_i, c_j, KB_i)
7: relKB[i] = getRelation(c_i, c_j, KB_i)
8: end if
9: end for
10: if(!isEmpty(relKB))
11: suggestedRel[i][j] = voteForRelation(relKB)
12: end if
13: end for
14: end for
15: buildOntology(suggestedRel)

Table 1
Identified semantic relations between the concepts of both ontologies.

	Agent	Person	Corporate Body	Concept	Place	Object	Artifact	Expression	Item	Work	Manifestation	Event
Agent	≡	⊂	??	⊥	⊥	⊂	≡	⊥	⊥	⊥	⊥	⊥
Person	⊃	≡	??	⊥	⊥	⊂	⊥	⊥	⊥	⊥	⊥	⊥
Corporate Body	??	??	≡	??	??	??	??	??	??	??	??	??
Concept	⊥	⊥	??	⊥	⊥	⊥	⊥	⊥	⊥	⊥	⊥	⊥
Place	⊥	⊥	??	⊥	≡	⊂	⊥	⊥	⊂	⊂	⊥	⊂
Object	⊃	⊃	??	⊥	⊥	≡	⊥	⊥	⊃	⊥	⊥	⊥
Artifact	⊥	⊥	??	⊥	⊥	⊂	≡	⊥	⊥	⊃	⊥	⊥
Expression	⊥	⊥	??	⊥	⊥	⊥	⊥	≡	⊥	⊥	≡	⊂
Item	⊥	⊥	??	⊥	⊥	⊂	⊥	⊥	≡	⊥	⊥	⊥
Work	⊥	⊥	??	⊥	⊥	⊂	⊂	⊥	⊥	≡	⊥	⊂
Manifestation	⊥	⊥	??	⊥	⊥	⊥	⊥	≡	⊥	⊥	≡	⊂
Event	⊥	⊥	??	⊥	⊃	⊥	⊥	⊃	⊥	⊃	⊃	≡

A nested loop is carried out to obtain the semantic relation between concept pairs (c_i, c_j) from the ontologies $\Omega 1$ and $\Omega 2$ according to their definitions in the knowledge bases. Then, for related concept pairs, the **voteForRelation** function (line 11) takes as input the relations from the knowledge bases and returns the suggested relations as output. At this step, decisions on the suggested relations are also made by a consensual decision from the knowledge bases. For example, at this step, we find the following relations between the concepts of the input ontologies:

1. Agent ≡ Agent, Agent ⊂ Group, Agent ⊂ Person
2. Corporate Body ≡ Corporate Body
3. Person ⊃ Agent, Person ⊂ Group, Person ⊃ Publisher, Person ≡ Person, Person ⊃Author
4. Group ⊃ Agent, Group ≡ Group, Group ⊃ Conference, Group ⊃ Organization, Group ⊃ University, Group ⊃ Publisher, Group ⊃ Person
5. Conference ⊂ Group, Conference ≡ Conference, Conference ⊂ Organization
6. Organization ⊂ Group, Organization ⊃ Conference, Organization ≡ Organization
7. University ⊂ Group, University ⊂ Organization, University ≡ University
8. Publisher ⊂ Group, Publisher ⊂ Organization, Publisher ⊂ Person, Publisher ≡ Publisher
9. Author ⊂ Person, Author ≡ Author

Then, the **buildOntology** function constructs the hierarchy of the merged ontology based on the computed semantic relations between the concepts of both ontologies.

4.3. Dealing with missing background knowledge

A potential issue is related to the unavailability of information regarding concepts and instances (also called *missing background knowledge*). We may indeed not be able to identify semantic relations during the merging process because one of the concepts or both do not exist in the knowledge bases. We therefore consider an enrichment process integrating name-based and coupled statistical and semantic techniques. The first technique allows determining concept or instance equivalence while the coupled technique makes it possible to highlight synonymy, hypernymy, hyponymy, meronymy, holonymy and instantiation relations.

4.3.1. Utilizing a name-based technique to highlight concept or instance equivalence

To obtain the set of equivalent elements between both ontologies, we use the Jaro-Winkler distance function [43] which is a simple and fast technique that measures the similarity between the strings of concepts and instances in both ontologies. The Jaro-Winkler similarity metric between two string s and t is given by:

$$\text{jaro} - \text{winkler}(s,t) = \frac{1}{3} \cdot \left(\frac{m}{|s|} + \frac{m}{|t|} + \frac{m - t'}{m} \right) \quad (3)$$

where

- s is the first string,
- t is the second string,
- M is the number of matching characters,
- t' is the number of transpositions.

Algorithm 3 demonstrates the process of finding equivalent concepts and instances between both ontologies $\Omega 1$ and $\Omega 2$. As mentioned above, these concepts and instances are missing in the exploited knowledge bases.

Algorithm 3. Name-based technique for finding concept/instance equivalence relations between $\Omega 1$ and $\Omega 2$

Input: vector of concepts or instances of Ω1 and Ω2 missing in the knowledge bases: *Ω1_miss* and *Ω2_miss*, threshold value *v*

Output: A structure *EqCI* of equivalent concepts/instances

```
1: int result
2: for i = 0; i < Ω1_miss.length; i++
3:   for j = 0; j < Ω2_miss.length; j++
4:     result = jaro-winkler(Ω1_miss[i], Ω2_miss[j])
5:     if(result > v) then
6:       add (Ω1_miss[i], Ω2_miss[j]) to EqCI
7:     end if
8:   end for
9: end for
```

The algorithm measures the similarity between strings corresponding to concepts/instances $\Omega 1$_miss[i] and $\Omega 2$_miss[j] that are missing in the knowledge bases (line 4). If the similarity measure is more than a threshold value v, then both are considered equivalent and added to the set of equivalent concepts/instances EqCI (lines 5 and 6).

4.3.2. Exploiting statistical and semantic information about missing background knowledge

We take advantage of the vast knowledge available on the Web by utilizing the Normalized Retrieval Distance (NRD) function to obtain the potential relations between missing background knowledge and other concepts and instances of the merged ontology. It is important to mention that information on the Web represents facts, people's opinions, and ideas. We know that this information is subjective and sometimes may represent incorrect semantic information. Therefore, in our approach we limit the subjectivity and reduce the chance of extracting incorrect semantic information by only measuring the semantic relatedness between the concepts and instances of the merged ontology and those which are missing in the knowledge bases. To do this, we obtain all the NRDs between each missing entity and other entities in the merged ontology and filter out the results by eliminating some distances that are below a certain threshold value. Using the NRD function enables us to measure the semantic relatedness between a missing entity in the knowledge bases and other entities in the merged ontology. However, we still do not know what types of semantic relations may hold between them. Therefore, we defined a list of patterns such as "is (the) same as", "is a(n)", "is a part of", "is an instance of"... to be exploited in order to derive the appropriate semantic relation. Synonymy, hypernymy, hyponymy, meronymy, holonymy and instantiation relations were considered. Both singular and plural forms of the terms were taken into account and patterns including negation operators such as "No _ is a(n) _" are excluded. These types of relations can be automatically obtained by utilizing the Semantic Relation Extractor (SRE) function. For each pair of semantically related terms, the SRE function returns the number of their hits by submitting each of the patterns to the search engines. Finally, the semantic relations between entity pairs are suggested based on the highest values returned from different search engines.

For example, to extract a hyponymy relation between two concepts, we build on the definition of this relation in [31]: a concept represented by the synset {x, x', ...} is said to be a hyponym of the concept represented by the synset {y, y', ...} if native speakers of English accept sentences constructed from such frames as "An x is a (kind of) y". Therefore, in order to extract the hyponymy/hypernymy relation between each missing concept and the concepts of the merged ontology, we submit the following query to several search engines: Q_i = "(*c_mis*) is a(n) (*c_in*)" where

- *c_mis* is a missing concept in the knowledge bases.
- *c_in* is a concept in the merged ontology.

As an illustration, to acquire the hypernyms of the concept "Corporate Body" we submit the following queries:

Q_1 = "Corporate Body is an Organization" which outputs 478, ($res_Q_1 = 478$).
Q_2 = "Corporate Body is a Publisher" which outputs 0, ($res_Q_2 = 0$).

Based on the obtained results of queries Q_i, the concepts in the merged ontology that co-occur more frequently with the missing concept are then suggested as its hypernyms. This step is based on using an automatic threshold value v to automatically decide which concepts among the list of retrieved concepts are suggested as hypernyms of the missing concept. The threshold value v is determined based on the following procedure:

1. By submitting queries Q_i we obtain the following results:
   $$res_total = [res_Q_1, res_Q_2, res_Q_3, \ldots, res_Q_n]$$
2. We find the maximum difference between the number of retrieved results as follows:
   $$v = maxDiff(res_Q_i, res_Q_j)$$
3. Based on v, a sublist of the list of concepts is then suggested. In our example, the returned suggestion is to consider the concept "Corporate Body" as a sub-concept of "Organization".

Algorithm 4 explains the procedure of deriving the semantics of missing concepts or instances in the knowledge bases through obtaining statistical information about them.

Algorithm 4. Utilizing statistical information to derive the semantics of missing concepts and instances

Input: vector of missing entities *e_mis*, vector of entities in the merged ontology *e_in* threshold value v
Output: A structure *suggestedRel* of suggested semantic relations between entities of *e_mis* and *e_in*

```
1:  string[][] suggestedRel;
2:  float[] result;
3:  float minNRD;
4:  int index_minNRD;
5:  bool noRel = true
6:  for i = 0; i < e_mis.length; i++
7:     minNRD = result[0] = NRD(e_mis [i], e_in [0])
8:     index_minNRD = 0
9:     for j = 0; j < e_in.length; j++
10:      result[j] = NRD(e_mis [i], e_in [j])
11:      if result[j] < minNRD
12:         minNRD = result[j]
13:         index_minNRD = j
14:      end if
15:      if(result[j] > v)
16:         suggestedRel[i][j] = SRE(e_mis [i], e_in [j])
17:         noRel = false
18:      Else
19:         suggestedRel[i][j] = null
20:      end for
21:      if noRel
22:         suggestedRel[i][index_minNRD] = "Related_To"
23:      end if
24:  end for
```

The NRD function (line 10) computes the semantic relatedness of *e_mis* [i] and *e_in* [j]. Then, the SRE function takes as input pairs of entities that have a normalized retrieval distance below the threshold value v (line 16). For each of the defined patterns, the function submits exact queries to the search engines to derive the semantic relations that may hold between *c_mis* [i] and *e_in* [j]. When the SRE function returns no relation, i.e. no result is returned for any of the patterns (and the boolean noRel is true in line 21), we define the semantic relation "Related_To" between the pair that has the strongest semantic relatedness (i.e. the minimum NRD value). This step generates a set of axioms $A_{external_resource} = \{(r, c_in_j, c_mis_k)\} \cup \{(r, c_mis_k, c_in_j)\}$ s.t. $r \in R_{merged} \cup \{Related_To\}$, c_mis_k is a missing concept in the knowledge bases (i.e. $c_mis_k \in C_1 \cup C_2 \wedge c_mis_k \notin \cup C_{KB_i}$) and $c_in_j \in C_{merged}$.

We then re-locate the missing concepts in the merged ontology and update its hierarchy. For example, it was suggested to consider the concept "Corporate Body" as a sub-concept of the concept "Organization". Therefore, the updated merged ontology Ω_{merged_update} will be defined as follows:

$\Omega_{merged_update} = \langle C_{merged}, R_{merged}, I_{merged_update}, A_{merged_update} \rangle$ where

- I_{merged_update} results from the enrichment of I_{merged} with novel instances found through this step and A_{merged_update} is the set of axioms defined by: $A_{merged_update} = (A_{merged}/\{(r, c_i, c_j)\}$ s.t. $c_i \notin \cup C_{KB_i}$ or $c_j \notin \cup C_{KB_i}) \cup A_{external_resource}$.

4.4. Knowledge base enrichment

We will attach the missing concepts to the generic knowledge base KB according to the semantic path(s) that originate from them in the merged ontology. Enriching the generic knowledge base with a new concept requires finding the appropriate position for locating it. To do so, we will check the path(s) that originate(s) from the missing concept in the merged ontology and map it/them to the proper path(s) in the generic knowledge base. Below we summarize the different cases that we consider for the enrichment.

Case 1: We have the concept *c_mis* (i.e. $c_mis \in C1 \cup C2 \wedge c_mis \notin C_{KB}$) in the merged ontology as a sub-concept of one direct super-concept $c \in C_{merged} \cap C_{KB}$. The concept c has only one sense in the generic knowledge base, i.e. there is only one semantic path that originates from c. In this case, we locate the missing concept *c_mis* under its direct super-concept c in the semantic path that originates from *c_mis* in the generic knowledge base. For example, the concept "Concept" in WordNet has only one sense. Therefore, any concept considered as a sub-concept of this concept will be directly located under it. Fig. 2 illustrates this example. This step enriches the set of axioms of the knowledge base such that $A_{KB} = A_{KB} \cup \{r, c_mis, c\}$ with $r \in R_{merged}$.
Case 2: We have the concept *c_mis* in the merged ontology as a sub-concept of one direct super-concept c. The concept c has more than one sense in the knowledge base, i.e. there is more than one semantic path originating from concept c.

In this case, based on the merged ontology and the knowledge base, we check the semantic similarity between the set of descendant concepts of concept c in the merged ontology and the list of descendant concepts of c in the knowledge base. Then, for the most similar semantic path(s) that originate from concept c, i.e. the semantic paths such that the number of similar concepts is maximum, we locate the missing concept *c_mis* as a sub-concept of the concept c.

Fig. 3 illustrates the case of enriching the knowledge base WordNet with the concept "Corporate Body". As we can see in

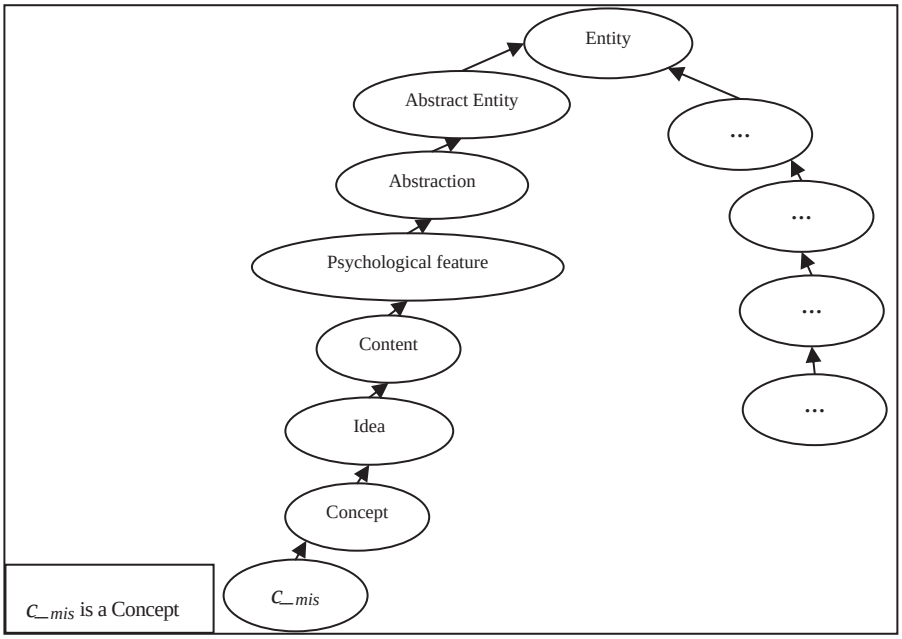

Fig. 2. Result of applying Case 1.

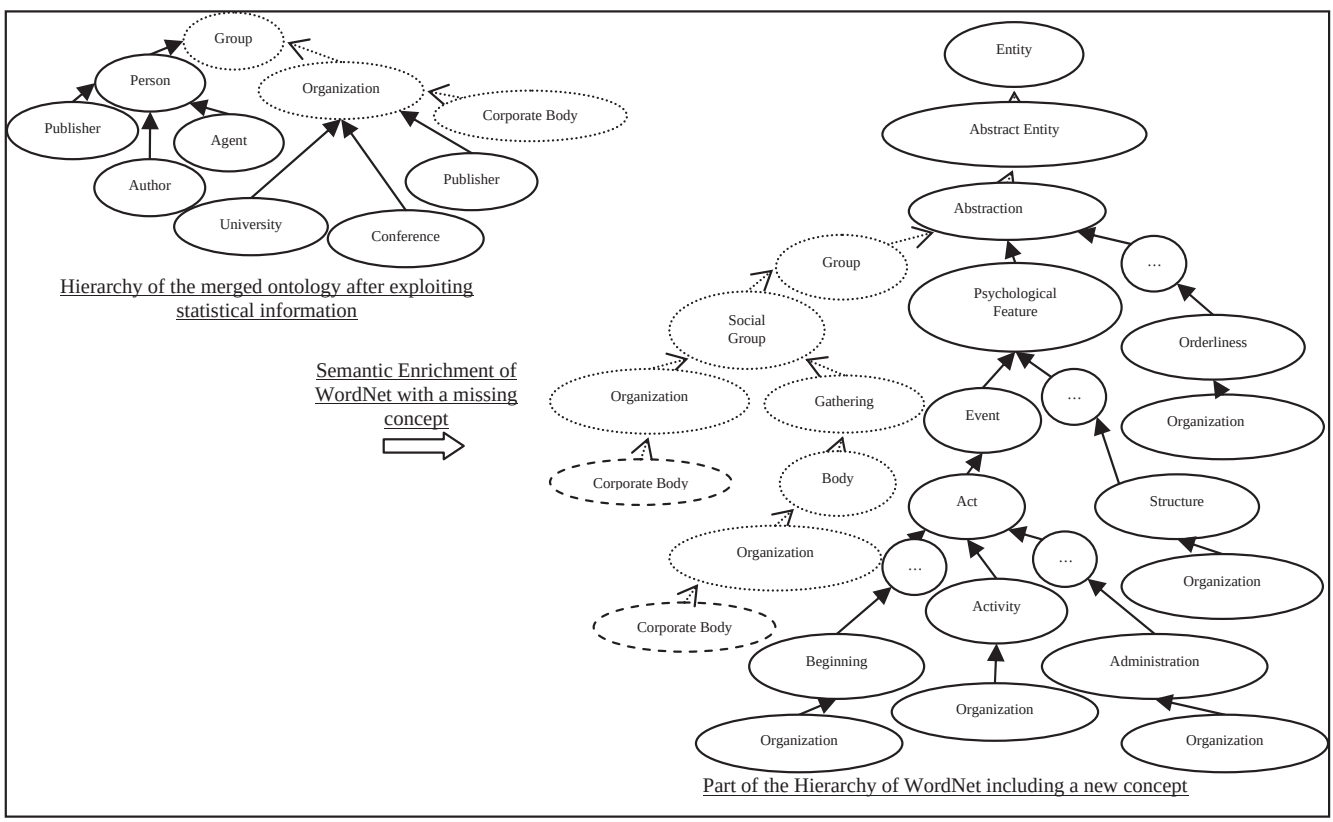

Fig. 3. Result of applying Case 2.

the merged ontology, this concept is a sub-concept of the concept "Organization". In WordNet, the concept "Organization" has more than one sense. Therefore, we need to find which sense among the several senses of this concept will be considered as a super-concept of "Corporate Body". This step can be done based on the comparison between the semantic path which originates from "Organization" in the merged ontology and those originating from "Organization" in WordNet. The semantic path(s) most similar to the one in the merged ontology will be considered in order to attach the concept "Corporate body" to them.

Let o denote the relation composition operator and $DKB_i = \{c_j, c_q \in C_{KB} | r_k o \ldots o r_l(c, c_j) \wedge r_m o \ldots o r_n(c, c_q) \wedge r_o o \ldots o r_p(c_j, c_q) \wedge r_k, \ldots,$

Table 2
Expert merging results produced by our system.

Ontologies	Expert merging	Correct	Not correct	Others
Google and Yahoo Web Directories (54 comparisons[a])	4	4	0	6
Product schemas (30 comparison)	4	4	0	0
Company profiles (9 comparison)	3	2	0	0
Simple catalogs (6 comparison)	3	3	0	0

[a] Is the number of times the elements of the ontologies are compared to each other.

Table 3
Precision/recall results.

Ontology	Cupid		COMA		S-Match		Our results	
	P	R	P	R	P	R	P	R
Simple catalogs (20)	0.44	0.36	0.62	0.66	1.00	1.00	1.00	1.00

$r_l, r_m, \ldots, r_n, r_o, \ldots, r_p \in R_{KB} \cup \{Id\}\}$ the set of descendants of concept c for each semantic path s_i in the knowledge base. Let DM = $\{c_i k, c_j \in C_{merged} | r_k\ o \ldots or_l(c, c_i) \wedge r_m o \ldots or_n(c, c_j) \wedge r_o o \ldots or_p(c_i, c_j) \wedge r_k, \ldots, r_l, r_m, \ldots, r_n, r_o, \ldots, r_p \in R_{merged} \cup \{Id\}\}$ the set of descendants of concept c in the merged ontology.

Let s be the semantic path corresponding to the set DKB such that $|DKB \cap DM| = \max(|DKB_i \cap DM|)$. In the case of a tie such as in Fig. 3, all the tied parties (i.e. the corresponding semantic paths) are considered. This step enriches the set of axioms of the knowledge base such that $A_{KB} = A_{KB} \cup \{r, c_mis, c\}$ with $c \in DKB$ and $r \in R_{merged}$.

Case 3: We have the concept *c_mis* in the merged ontology which is a sub-concept of more than one direct super-concept. In this case, we check whether either Case 1 or Case 2 applies for each direct super-concept of the missing concept *c_mis*.

5. Experimental results

We propose a prototypal implementation of our theoretical solution in Java and experiments are performed on a PC with dual-core CPU (3 GHz) and 4 GB of RAM. The operating system is OpenSuse 11.1. The exploited knowledge bases are WordNet [14], OpenCyc [17], and Yago [18].

5.1. Experiments using expert merging results

In these experiments, we used the expert merging results[3] which were manually produced in [17,41]. We used four pairings of ontologies: Parts of Google and Yahoo Web directories, Product Schemas, Simple Catalogs, and Company profiles. The results produced by our system are classified into: (i) correct: where the produced semantic relations between the concepts of both ontologies are correct and exist in the expert merging results, (ii) incorrect: where the produced semantic relations are not correct and (iii) others: where the produced semantic relations are correct but do not exist in the expert merging results.

In Table 2, we compare the results produced by our system to the expert merging results. Then, we compare our results to three of the state-of-the-art syntactic-based and semantic-based systems: S-Match [17], Cupid [30], and COMA [13].

As shown in Table 2, our system produced the semantic relations that hold between the elements of the ontologies in all of the tests except for the Company profiles test. This is due to the problem of missing background knowledge in the exploited knowledge bases. For the first test, our system produced six other relations that do not exist in the expert merging results. Indeed, all of these relations are important mainly if we want to integrate the merged ontology to other generic ontologies. In this context, these relations can improve the quality and coverage of the integrated ontology.

In Table 3, we show our comparative results to the above mentioned systems. We use precision and recall performance measures to measure the performance of our system. These measures are defined as follows. Precision (P) provides the percentage of the correct discovered semantic correspondences considering all discovered semantic correspondences between concepts. Recall (R) provides the percentage of the correct discovered semantic correspondences considering all correct semantic correspondences between concepts.

The number of concepts in the test ontology is 20 concepts (4 concepts in the first ontology and 5 concepts in the second ontology). As shown in Table 3, our system shows higher precision and recall than both the COMA and Cupid systems and is furthermore on par with the S-Match system. However, for heavyweight ontologies, the S-Match system will potentially face computational issues as it utilizes graph-matching techniques that build in-memory structures to merge ontologies.

5.2. Experiments Using the OAEI [22] Benchmarks

In the experiments involving real-world ontologies, we first used the Ontology Alignment Evaluation Initiative (OAEI) 2009 datasets (test series 3xx). Then, we considered three mainstream heavyweight ontologies (GEMET [23], NAL [25] and AGROVOC [24]) to carry out the second part of the experiments.

5.2.1. Experiments using the real-world ontologies of the benchmark test series 3xx

These benchmark test series are based on one particular ontology dedicated to the domain of bibliography and a number of alternative ontologies of the same domain for which merging results (in the form of semantic correspondences between concepts) are provided. Some information has been discarded from the ontologies in the datasets in order to evaluate how the algorithm behaves when this information is lacking. A detailed summary on what has been retracted from each of the ontologies can be found in ("http://oaei.ontologymatching.org/,"). The ontology corresponding to test 301 (i.e. BibTeX/MIT) is a bibliographic ontology that is widely used in the domain. The ontology corresponding to test 302 (i.e. BibTeX/UMBC) is very similar to the previous one, even closer to the original BibTeX, with different extensions and naming conventions. The ontology corresponding to test 303 (i.e. Karlsruhe) is used in the Ontoweb portal and defines more items than the items defined in the reference ontology. The ontology corresponding to test 304 (i.e. INRIA) is itself close to the reference ontology. It is important to note that these ontologies are developed by experts and are used in real-world applications.

The following experiments show the effectiveness of the first two strategies that we utilize in the framework, i.e. semantic-based

[3] http://dit.unitn.it/~accord/Experimentaldesing.html.

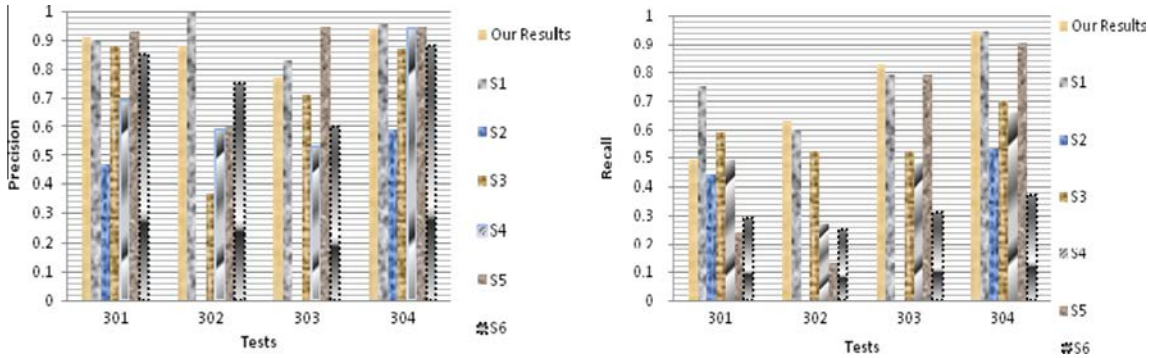

Fig. 4. Precision/recall results for the ontologies of the benchmark test series 3xx.

Table 4
Experiments using heavyweight real-world ontologies.

Task	# of Semantic correspondences
GEMET-AGROVOC	
Chemistry-Precision	14 out of 14
Geography-Precision	23 out of 23
Geography-Recall	87 out of 87
Agriculture-Recall	61 out of 61
Misc-Precision	28 out of 28
Tax-Precision	21 out of 21
Risk-Precision	21 out of 21
Nat-Precision	35 out of 35
NAL-AGROVOC	
Chemistry-Precision	141 out of 141
Geography-Precision	58 out of 58
Misc-Precision	231 out of 231
Tax-Precision	10 out of 10
Anim-Recall	10 out of 10
Rod-Recall	24 out of 24
Oaks-Recall	38 out of 38
Eur-Recall	62 out of 62
Geography-Recall	58 out of 58
GEMET-NAL	
Chemistry-Precision	30 out of 30
Geography-Precision	17 out of 17
Misc-Precision	29 out of 29
Tax-Precision	15 out of 15
Nat-Precision	23 out of 23
Risk-Precision	30 out of 30
Agriculture-Recall	61 out of 61
Geography-Recall	77 out of 77

and name-based. We compare our results with state-of-the-art systems classified according to the different strategies and techniques they use as follows:

1. *Multi-layer Ontology Merging Systems*: these systems divide the ontology merging task into three different layers. In the first layer, concept features such as labels, comments, and instances are compared using a variety of syntactic and lexical methods. In the second layer, the structural properties of the ontologies are used. The third layer combines the results from the first two layers to produce unique merging results. Examples of these systems are state-of-the-art systems S1 [9] and S2 [4].
2. *String-based and Structure-based Ontology Merging systems*: only these two strategies are used to align the ontologies. First, the string-based technique is employed to measure the similarity between the strings of concepts and properties of the ontologies. Then, the ontology class hierarchy is used in the structure-based technique to obtain equivalent concepts in both ontologies. Examples of these systems are state-of-the-art systems S3 [35], S4 [36], S5 [44] and S6 [20].

Precision and recall comparative results for test series 3xx are shown in Fig. 4. In terms of precision, we report improvements over S_2, S_3, S_4, S_5 and S_6 (statistically significant over S_2, S_4 and S_6 with a paired-samples *t*-test although considering 4 tests only). This translates the fact *that our system is particularly able to dismiss* any incorrect semantic correspondences among those found. In terms of recall, we also report improvements over S_2, S_3, S_4, S_5 and S_6 (statistically significant with a paired-samples *t*-test over S_2, S_3, S_4 and S_6). This highlights the propensity of the system to find all the correct semantic correspondences. Let us also note that our system obtained comparable precision and recall results with S_1. However, the latter only attempts to find alignments based on concept equivalence and does not consider other types of semantic relations. Also, contrarily to our system (as shown in Section 5.2.2), S_1 was not experimentally shown to operate with heavyweight ontologies.

5.2.2. Experiments using heavyweight real-world ontologies

In these experiments we find correspondences between the concepts and instances of three heavyweight real-world ontologies (GEMET, AGROVOC, and NAL). Details on these ontologies are listed below:

1. *GEMET*: The General Multilingual Environmental Thesaurus has been developed as an indexing, retrieval and control tool for the European Topic Centre on Catalogue of Data Sources and the European Environment Agency. GEMET was conceived as a general thesaurus, aimed to define a common general language, a core of general terminology for the environment. The number of concepts in this ontology is 5280.
2. *NAL*: The NAL Agricultural Thesaurus was originally prepared by staff of the National Agricultural Library to meet the needs of the United States Department of Agriculture and Agricultural Research Service. Its first edition was published on January 1, 2002. The thesaurus is primarily used for indexing and for improving retrieval of agricultural information. The number of concepts in this ontology is 42326.
3. *AGROVOC*: It is a multilingual structured thesaurus of all subject fields in agriculture, forestry, fisheries, food and domains related to environment. It consists of words or expressions (terms) in different languages and organized in relationships (e.g. broader, narrower and related) used to identify or search resources. Its main role is to standardize the indexing process in order to make searching simpler and more efficient, and to provide the user with the most relevant resources. The number of concepts in this ontology is 39814.

To compute precision and recall, we used the official golden standard merging results (in the form of semantic correspondences between concepts) [26] that are provided by the OAEI 2007

environment task organizers. These sample correspondences are classified into different semantic domains such as chemistry, geography and agriculture. We used these sample results to compute the precision and recall of our system for each semantic domain. A comparison between the results of our system and the golden standard reference results is shown in Table 4.

As shown in Table 4, for all of the merging tasks, our system produced the same number of equivalent concepts as in the reference results provided in the golden standard. Experiments using the same dataset were conducted in [45] and the produced results were the same as our results. However, our system can be distinguished from the system in [45] as we utilize a coupled statistical and semantic technique using the NRD function to address the problem of missing background knowledge in the exploited knowledge bases while they only propose a name-based technique to address the issue. In addition, we consider relations other than "Equivalence" between the concepts and instances of the ontologies.

6. Conclusions and future work

We have presented a fully-automated framework for merging domain-specific ontologies through exploiting semantic, name, and statistical based techniques. First, the system employs the semantics-based technique to merge heterogeneous ontologies through finding semantic relations that hold between their concepts based on their definitions in the exploited external knowledge bases. In case where the semantic relations cannot be identified (because one of the concepts or both do not exist in the knowledge bases), the system utilizes a name-based technique to find corresponding entities (only equivalent entities in this case) in the ontologies. A coupled statistical and semantic technique is further utilized to derive additional semantic relations between missing concepts and concepts in the merged ontology. These techniques were also used in the case of missing information regarding instances. We moreover studied how these additional discovered relations could be used to enrich the background knowledge in used knowledge bases. We extensively tested the proposed framework using several publically available datasets. We further compared the produced results by our system to the results produced by state-of-the-art ontology alignment and merging systems. Indeed, experimental results demonstrated the effectiveness of the employed techniques in identifying semantic correspondences between the entities of different ontologies. Importantly, we showed that the proposed framework effectively handled heavyweight ontologies that consist of thousands of entities such as concepts and instances.

As a future work, we plan to employ a segmentation mechanism that divides heavyweight real-world ontologies into smaller sub-ontologies. This process allows us to optimize the merging process through the incorporation of the structure-based ontology alignment and merging approach. We will study the significance of employing this approach and its effects on the quality of the produced alignment and merging results. In addition, we plan to integrate the used knowledge bases into a single knowledge base, which can be later used as a single repository of semantic information about multiple and heterogeneous domains.

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